

Jen Yang Koh

Product Engineer

Product engineer across the full product loop — user research, spec, build, ship, iterate.

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EXPERIENCE

Software Engineer | Latium Technologies | Sep 2022 - Present

- Owned a customer-facing web + mobile product as one of two full-stack engineers on a cross-functional team — the daily-use platform for site teams across 100+ active construction sites
- Halved camera calibration from 4 hours to 2 hours per site after hand-aiming cameras myself — a least-squares plane fit with RBF interpolation computes tilt from calibration data
- Shipped a concrete-curing sensor integration to field launch — ran on-site discovery with curing crews, aligned with stakeholders at the sensor vendor, and delivered temperature + humidity monitoring to field teams
- Launched the mobile app from scratch in .NET MAUI, displacing a shelved legacy app since the 2024 App Store debut — now used daily with icon-first navigation for low-literacy contexts
- Delivered camera livestreaming and PTZ control in the web client (200+ cameras, ~5s latency for monitoring) — spanning Angular frontend, API, and database, eliminating on-site physical access for camera operation
- Cut sensor-to-camera trigger latency from 5-30s to 1-2s for time-sensitive site security events — split telemetry into priority/non-priority queues so alarms beat routine traffic (Event Hub auto-scaling, device-based partitioning)

Freelance Web Developer | Eagles Communications, Singapore | Mar 2026 - Present

- Rebuilt eagles.org.sg for a Singapore nonprofit — migrated from legacy WordPress to Next.js + Prismic CMS with web-native newsletters, shipped to production. eagles.org.sg (<https://eagles.org.sg>)

Software Developer | PCL Construction | Jun 2021 - Sep 2022

- Developed platform-performance monitoring tooling as sole developer (React, .NET Core) for the JSI IoT platform — the same product I later led end-to-end at Latium
- Extended the JSI platform with follow-up reminder notifications (email, SMS, Teams) that kept site teams from missing time-sensitive alarms

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C#, Golang, SQL (MS SQL, PostgreSQL), HTML/CSS, Python

Frameworks & Libraries: React, Next.js, .NET, .NET MAUI, Angular, React Native, TailwindCSS, RxJS

Cloud & DevOps: Microsoft Azure (Functions, App Service, Service Bus, Event Hub, IoT Hub, Key Vault, Blob, Queue, Table), AWS (EC2, S3), Cloudflare R2, Docker, Azure Pipelines, Git

Architectures & Patterns: Event-Driven Architecture, Serverless, IoT Integration, RESTful APIs, CI/CD, MVVM

Developer Tools: VS Code, Visual Studio, Git

PROJECTS

DouDou | React, TailwindCSS, Supabase, S3 | Oct 2024 - Present

- Built a real-time photo voting app with multi-phase session workflow (join → submit → vote → award)
- Used by 50 participants in a single live session, each submitting a photo

EDUCATION

University of Alberta | Edmonton, Alberta, Canada

Bachelor of Science in Computer Science, Minor in Psychology